

Open Take-Home - AI-Driven Detection of Location Spoofing

Role fit: Software Engineering Intern (AI Systems)

Effort guideline: 4–5 focused hours (max 8)

Return window: Within 72 hours of receiving this prompt

1) Problem

Our SDK receives transaction-like events from mobile devices (iOS/Android). We need to detect location spoofing. Location spoofing is when a user or program makes a mobile device appear to be in a different place—or moving differently—than it really is. This can distort in-app experiences, bypass geo-based rules, or enable fraud. Your job is to design data and build a detector that flags events where the reported location context is likely untrue.

You will independently use **AI tools** to:

1. Design & synthesize a realistic dataset (no data is provided). Feel free to choose one of the operating systems (iOS or Android).
2. Build a compact detection system (transparent rules + a small ML model).
3. Perform a self-evaluation with honest analysis and trade-offs.

2) Data

You will propose a schema, generate synthetic data, and document it.

2.1 Required scope

- **Two splits:** **train** (labeled) and **test** (unlabeled for scoring—though you'll know the ground truth when generating).
- **Scale:** Aim for ~10k–25k rows train and ~3k–8k rows test (choose a size you can handle).
- **Label prevalence:** Spoof positives within **10%–30%** in train; report your exact rate.

- **Synthetic only:** No scraping or real PII.

2.2 SDK-collectable features (AI-driven expansion; required)

- Treat your schema as what a **mobile SDK could plausibly collect** (subject to platform policies/permissions).
- Add fields beyond basic lat/lon/timestamp (you choose what to include and justify)

For every added field, document: *how an SDK could obtain/infer it, why it helps detect spoofing, and expected value range/type.*

3) What to build

3.1 Brief plan

- Your spoofing hypotheses, chosen schema, and **where AI will help** (brainstorming, code generation, rule tuning, error analysis).

3.2 Rules/heuristics baseline

- Using AI to create heuristics baseline.
- Examples for rules:
 - Teleportation / impossible speed between consecutive points.
 - IP/Geo mismatch (e.g., lat/lon in CA, ip address in the US).

3.3 ML model (compact)

- Build a classification model to detect location spoofing
- Use features derived from your expanded schema.
- Output `spoof_score_ml` and `spoof_flag_ml`; justify your decision threshold (precision target, cost trade-off, or PR analysis).

3.4 AI/LLM augmentation

- **Rule proposer:** Given schema + examples, AI suggests candidate rules; you evaluate and keep/tune the good ones.
- **Weak labeling:** AI reads textual notes (you can synthesize short **_note* strings) to produce weak labels for a slice; compare vs ground truth.
- **Explainer:** For a flagged event/session, produce a short natural-language justification a reviewer could trust.

(If you can't call an external API, include prompts and a deterministic "mocked" path so we can reproduce.)

4) Self-evaluation (you must design & run this)

4.1 Metrics

For example:

- Report **Precision, Recall, F1** with **spoof as positive**.
- Plot **PR curve** and explain your chosen operating point.
- Provide a tiny **ablation**: Rules-only vs ML-only vs Hybrid.

4.2 Error analysis

- Discuss **5 false positives** and **5 false negatives**: what patterns caused them, how rules/thresholds could be tuned, and where AI helped your investigation.

5) Deliverables (zip or repo)

```
/submission
/data                # your synthetic data
  train.csv
  test.csv
/src
  generate_data.py    # or a notebook + a simple runner script
  rules_baseline.py
  model_train_eval.py
```

```
    ai_helper.py          # rule proposer / explainer / weak labeler
(can be mocked)
    README.md             # one-command run steps
    DATACARD.md           # schema, ranges, spoof taxonomy,
distributions, limitations
    AI_USAGE.md           # which AI tools, prompts, what was
accepted/edited
    EVAL_REPORT.md        # metrics, PR curve, ablation, FP/FN
analysis, chosen threshold
    design.pdf (or .md)   # ≤1 page:
ingestion→features→rules→ML→AI→decision
    results.json          # {event_id, spoof_score, spoof_flag,
explanation?} for test
    requirements.txt
```

6) Helpful AI prompts (optional, adapt as you like)

- “Propose 15 SDK-collectable mobile telemetry fields for GPS spoofing detection; for each, give iOS/Android feasibility and a realistic numeric/categorical range.”
- “Draft Python to synthesize per-device tracks with walking/driving segments, realistic jitter, and configurable spoof injection for teleportation, IP/Geo mismatch, and time drift.”
- “Given this schema and three positive/negative examples, propose five interpretable rules with threshold candidates and note likely false positives.”
- “Explain in 2–3 sentences why this specific event is likely spoofed using fields A, B, C.”

7) Submission notes

Be explicit about how you used AI: which tools, which prompts, what you accepted vs edited, and why. Modest metrics are fine—**clarity and reasoning matter more than raw F1**.